

Data Mining and Big Data Anlysis
数据挖掘及大数据分析

9. Outlier Detection 离群点检测

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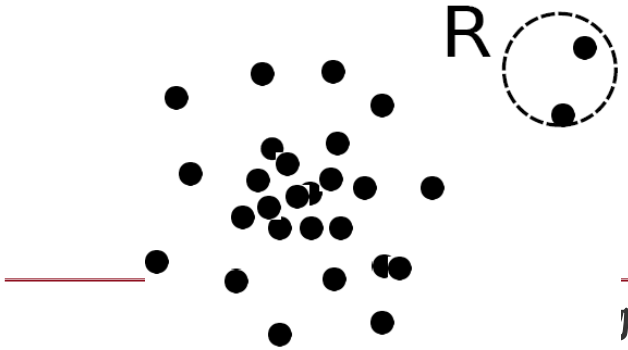
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Chapter 9. Outlier Detection

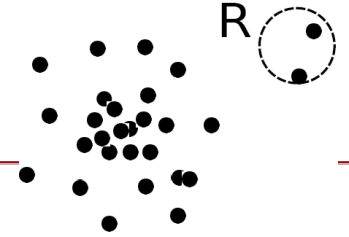
Outlier and Outlier Analysis	**离群点和离群点分析**
Outlier Detection Methods	离群点检测方法
Statistical Approaches	统计学方法
Proximity-Base Approaches	基于邻近性的方法
Clustering-Base Approaches	基于聚类的方法
Classification Approaches	基于分类的方法
Mining Contextual and Collective Outliers	挖掘情境离群点和集体利群点
Outlier Detection in High Dimensional Data	高维数据中的离群点检测
Summary	小结

What Are Outliers?

- **Outlier**: A data object that deviates significantly from the normal objects as if it were generated by a different mechanism
  - Ex.: Unusual credit card purchase, sports: Michael Jordon, ...
- **Outliers are interesting**: It violates the mechanism that generates the normal data
- **Outliers离群点 are different from the noise data噪声数据**
  - Noise is random error or variance不同 in a measured variable被测变量的随机误差或方差，如：比往常多买一杯咖啡……
  - Noise should be removed before outlier detection一般不是令人感兴趣的
- **Outlier detection离群点检测 vs. *novelty detection新颖性检测***
  - 离群点检测还与演变数据集上的新颖性检测相关，例如，通过检测新内容不断出现的社会媒体网站，新颖性检测可以及时地识别新的主题和趋势。新主题最初可能以离群点形式出现。在某种程度上，离群点检测与新颖性检测在建模和方法上都有许多相似之处。 early stage, outlier;
  - 两者的关键区别是，在新颖性检测时，一旦新主题被证实，则通常把它们合并到正常行为的模型中，这样接踵而来的实例不再被视为离群点。but later merged into the model
- Applications:
  - Credit card fraud detection
  - Telecom fraud detection
  - Customer segmentation分割
  - Medical analysis



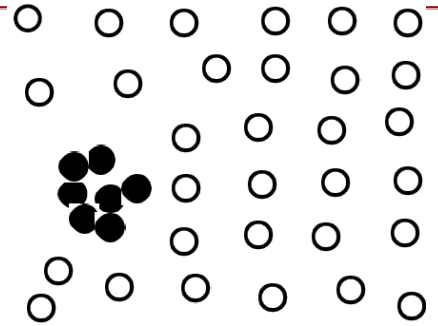
Types of Outliers (I)



- Three kinds: *global, contextual* and *collective* outliers
- Global outlier全局离群点(or point anomaly点异常)
  - Object is O_g if it significantly deviates from the rest of the data set
    - 一台计算机的通信行为非常不同于正常模式。例如，在短时间内大量的包被广播则该行为可以看做一个全局离群点，而对应的计算机可能是黑客入侵的受害者
    - 在交易审计系统中，不遵守常规的交易可能被视为全局离群点，并且应该搁置，以便进一步考察
  - Issue: 针对所考虑的应用Find an appropriate measurement of deviation偏离度量
- Contextual outlier情境离群点(or conditional outlier条件离群点)
  - Object is O_c if it deviates significantly based on a selected context特定情境
  - Ex. 北京下雪: outlier? (depending on summer or winter?)
  - Attributes of data objects should be divided into two groups
    - Contextual attributes情境属性: defines the context, e.g., time & location
    - Behavioral attributes行为属性: characteristics of the object, used in outlier evaluation, e.g., temperature
  - Can be viewed as a generalization of *local outliers局部离群点的推广*局部离群点是基于密度的离群点检测方法引进的概念。数据集中的一个对象是局部离群点，如果它的密度显著地偏离它所在的局部区域的密度whose density significantly deviates from its local area
  - 全局离群点检测可以看做情境离群点检测的特例，其中情境属性集为空
    - 全局离群点检测使用整个数据集作为情境
  - 情境离群点分析为用户提供了灵活性，因为用户可以在不同的情境下考察离群点，这在许多应用中都是非常期望的。
  - Issue: How to define or formulate meaningful context?

Types of Outliers (II)

- Collective Outliers集体离群点
  - A subset of data objects *collectively* deviate significantly from the whole data set, even if the individual data objects may not be outliers
  - Applications: E.g., *intrusion detection*:
    - When a number of computers keep sending denial-of-service packages to each other 相互发送拒绝服务包
  - Detection of collective outliers
    - Consider not only behavior of individual objects, but also that of groups of objects
    - Need to have the background knowledge on the relationship among data objects, such as a distance or similarity measure on objects.



Collective Outlier

- A data set may have multiple types of outlier
- One object may belong to more than one type of outlier

Challenges of Outlier Detection

- Modeling normal objects and outliers properly
  - Hard to enumerate all possible normal behaviors in an application
  - The border between normal and outlier objects is often a gray area
- Application-specific outlier detection
  - Choice of distance measure among objects and the model of relationship among objects are often application-dependent
  - E.g., clinic data诊所数据: a small deviation could be an outlier小偏离可能是重要的，足以证实离群点; while in marketing analysis, larger fluctuations在市场分析中，对象通常有很大的波动，因此需要显著的偏差才能证实离群点
  - 离群点检测高度依赖于应用类型，使得不可能开发通用的离群点检测方法。相反，必须开发针对具体应用的离群点检测方法
- Handling noise in outlier detection
  - Noise may distort扭曲 the normal objects and blur模糊 the distinction between normal objects and outliers. It may help hide outliers and reduce the effectiveness of outlier detection离群点像“伪装的”噪声点，错误地将噪声点识别成离群点
- Understandability
  - Understand why these are outliers: Justification of the detection在许多应用中，用户可能不仅要检测离群点，而且要知道被检测到的点为何是离群点。为了满足可理解性要求，离群点检测方法必须提供某种检测理由。
  - 例如，使用统计学方法，基于对象被大多数数据的相同机制产生的似然性，说明该对象是离群点的可能性。似然越小，该对象越不太可能被相同的机制产生，并且越可能是离群点

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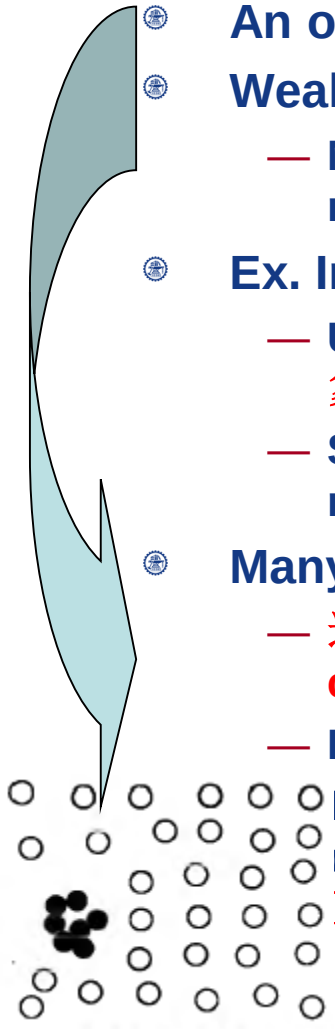
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Outlier Detection I: Supervised Methods

- Two ways to categorize outlier detection methods:
  - Based on whether user-*labeled* examples of outliers can be obtained:
    - Supervised, semi-supervised vs. unsupervised methods
  - Based on *assumptions about normal data and outliers*正常对象和离群点的假定:
    - Statistical, proximity-based, and clustering-based methods
- Outlier Detection I: Supervised Methods
  - Modeling outlier detection as a classification problem
    - Samples examined by domain experts used for training & testing
    - 监督方法对数据的正常性和异常性建模：领域专家考察并标记基础数据的一个样本，然后离群点检测可以用分类问题建模，任务是学习一个可以识别离群点的分类器。样本用于训练和检验
  - Methods for Learning a classifier for outlier detection effectively:
    - Model normal objects & report those not matching the model as outliers, or
    - Model outliers and treat those not matching the model as normal
  - Challenges
    - Imbalanced classes两个类（正常对象和离群点）不平衡, i.e., outliers are rare会限制构建的分类器的能力: 可以使用处理不平衡类的方法Boost the outlier class，如离群点过抽样（即复制） and make up some artificial outliers
    - Catch as many outliers as possible, i.e., recall is more important than accuracy (i.e., not mislabeling normal objects as outliers)

Outlier Detection II: Unsupervised Methods

- **Assume** the normal objects are somewhat ``clustered'' into multiple groups, each having some distinct features无监督的离群点检测方法暗中假定：正常对象在某种程度上是“聚类的”
- An outlier is expected to be far away from any groups of normal objects
- Weakness: **Cannot detect collective outlier effectively**
  - Normal objects may not share any strong patterns, but the collective outliers may share high similarity in a small area
- Ex. In some intrusion or virus detection, normal activities are **diverse**发散
  - Unsupervised methods may have a high **false positive rate**假正例率（把正常对象误标记为离群点） but still miss many real outliers.
  - Supervised methods can be more effective, e.g., identify attacking some key resources（入侵或病毒都攻击目标系统的关键资源）
- Many clustering methods can be adapted for unsupervised methods
  - 这些方法的核心思想是先**Find clusters,**然后 **then outliers: not belonging to any cluster**
  - Problem 1: Hard to distinguish noise from outliers
  - Problem 2: Costly since first clustering: but far less outliers than normal objects在可以触及实际内容（即离群点）之前必须先处理大量非目标数据实体（即正常对象）可能不那么吸引人
- Newer methods: **tackle outliers directly**，而不必显式和完全找出簇



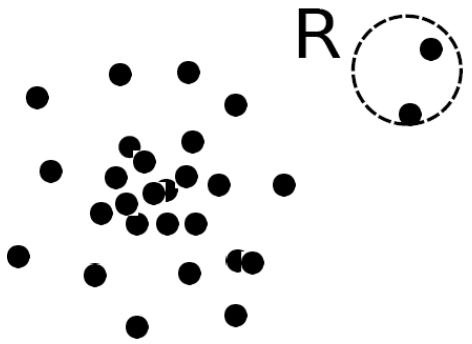
Outlier Detection III: Semi-Supervised Methods

- Situation: In many applications, the number of labeled data is often small: Labels could be on outliers only, normal objects only, or both
- Semi-supervised outlier detection半监督方法: Regarded as applications of semi-supervised learning
- If some labeled normal objects are available
  - Use the labeled examples and the proximate unlabeled objects与邻近的非标记对象 to train a model for normal objects
  - Those not fitting the model of normal objects are detected as outliers
- If only some labeled outliers are available, a small number of labeled outliers many not cover the possible outliers well
  - To improve the quality of outlier detection, one can get help from models for normal objects learned from unsupervised methods

Outlier Detection (1): Statistical Methods
统计学方法（基于模型的方法）

- **Statistical methods (also known as model-based methods) assume that the normal data follow some statistical model (a stochastic model 随机模型)**
  - **The data not following the model are outliers.**

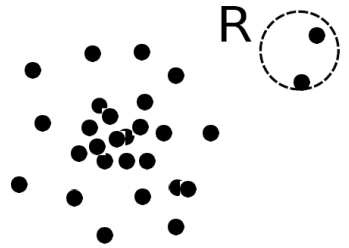
- Example (right figure): First **use Gaussian distribution to model the normal data**
  - 除区域R外的点都拟合一个高斯分布g_D, 其中对于数据空间的每个位置x， $g_D(x)$给出x上的概率密度
  - For each object y in region R, estimate $g_D(y)$, the probability of y fits the Gaussian distribution
  - If $g_D(y)$ is very low, y is unlikely generated by the Gaussian model, thus an outlier



- Effectiveness of statistical methods: highly depends on whether the assumption of statistical model holds in the real data
- There are rich alternatives to use various statistical models
  - E.g., parametric vs. non-parametric

Outlier Detection (2): Proximity-Based Methods
基于邻近性的方法

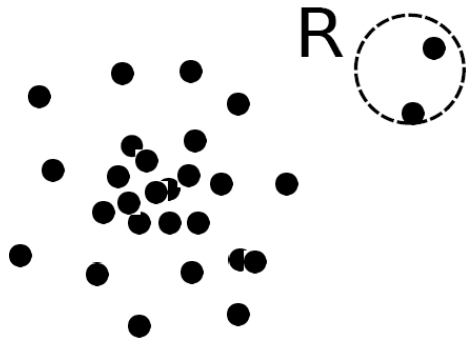
- **An object is an outlier if the nearest neighbors of the object are far away, i.e., the** proximity **of the object is** significantly deviates **from the proximity of most of the other objects in the same data set**



  - **Example (right figure): Model the proximity of an object using its 3 nearest neighbors**
    - **Objects in region R are substantially different from other objects in the data set on 第二和第三个最近邻.**
    - **Thus the objects in R are outliers**
  - The effectiveness of proximity-based methods highly relies on the **proximity measure邻近性（或距离）度量**.
  - In some applications, proximity or distance measures cannot be obtained easily.
    - Two major types of proximity-based outlier detection
      - Distance-based vs. density-based
  - Often have a difficulty in finding a group of outliers which stay close to each other如果离群点相互靠近，则本方法常常很难检测离群点的组群

Outlier Detection (3): Clustering-Based Methods
基于聚类的方法

- 假定**Normal data belong to large and dense clusters, whereas outliers belong to small or sparse clusters, or do not belong to any clusters**



- Example (right figure): two clusters
  - All points not in R form a large cluster
  - The two points in R form a tiny cluster, thus are outliers
- Since there are many clustering methods, there are many clustering-based outlier detection methods as well
- Clustering is expensive: straightforward adaption of a **clustering method for outlier detection can be costly and does not scale up well for large data sets**

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Statistical Approaches
统计学方法

- Statistical approaches assume that the objects in a data set are generated by a stochastic process随机过程 (a generative model生成模型)
- Idea: learn a generative model fitting the given data set, and then identify the objects in low probability regions低概率区域 of the model as outliers
- 学习生成模型的方法Methods are divided into two categories: *parametric* vs. *non-parametric*
- Parametric method
  - Assumes that the normal data is generated by a parametric distribution with parameter θ
  - The probability density function概率密度函数 of the parametric distribution $f(x, \theta)$ gives the probability that object x is generated by the distribution
  - The smaller this value, the more likely x is an outlier
- Non-parametric method
  - Not assume an a-priori statistical model and determine the model from the input data不假定先验统计模型，而是试图从输入数据确定模型
  - Not completely parameter free but consider the number and nature of the parameters are flexible and not fixed in advance大多数非参数方法并不假定模型是完全无参的（完全无参假定将使得从数据学习模型是不可能的），相反，非参数方法通常假定参数的个数和性质都是灵活的，不预先确定
  - Examples: histogram直方图 and kernel density estimation核密度估计

Parametric Methods I: Detection Univariate Outliers Based on Normal Distribution

- **Univariate data一元数据: A data set involving only one attribute or variable**
- **方法原理Often assume that data are generated from a normal distribution, learn the parameters from the input data, and identify the points with low probability as outliers**
- **Ex: Avg. temp.: {24.0, 28.9, 28.9, 29.0, 29.1, 29.1, 29.2, 29.2, 29.3, 29.4}**
  - 通过假设上述温度数据符合正态分布来检测离群点
  - **Use the maximum likelihood method最大似然方法，即最大化对数似然函数 to estimate μ and σ**

$$\ln \mathcal{L}(\mu, \sigma^2) = \sum_{i=1}^{n} \ln f(x_i|(\mu, \sigma^2)) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

- Taking derivatives求导 with respect to μ and σ^2, we derive the following **maximum likelihood estimates最大似然估计**

$$\hat{\mu} = \overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

- **For the above data with n = 10, we have** $\hat{\mu} = 28.61 \quad \hat{\sigma} = \sqrt{2.29} = 1.51$
- **最大偏离值 24.0偏离均值4.61，在正态分布中(24 – 28.61) /1.51 = – 3.04 < –3, 24 is an outlier since** $\mu \pm 3\sigma$ region contains 99.7% data

24被该正态分布产生的概率小于15%，被识别为离群点

Parametric Methods I: The Grubb's Test
Grubb检验（最大标准残差检验）

- Univariate outlier detection: The Grubb's test (maximum normed residual test) – another statistical method under normal distribution
  - For each object x in a data set, compute its z-score z分数: x is an outlier if

$$z = \frac{|x - \bar{x}|}{s}$$

标准差 均值

$$z \geq \frac{N-1}{\sqrt{N}} \sqrt{\frac{t^2_{\alpha/(2N),N-2}}{N-2+t^2_{\alpha/(2N),N-2}}}$$

where $t^2_{\alpha/(2N),N-2}$ s the value taken by a t-distributiont-分布值 at a significance level显著水平 of α/(2N), and N is the # of objects in the data set

Parametric Methods II: Detection of Multivariate Outliers
多元离群点检测

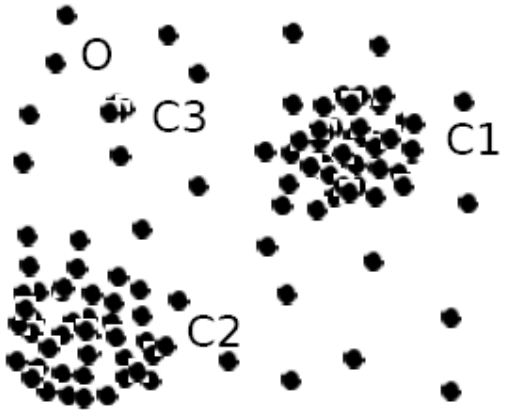
- Multivariate data: A data set involving two or more attributes or variables
- Transform the multivariate outlier detection task into a univariate outlier detection problem
- Method 1. Compute Mahalaobis distance使用马哈拉诺比斯距离识别离群点
  - Let $\bar{o}$ be the mean vector均值向量 for a multivariate data set. Mahalaobis distance for an object o to $\bar{o}$ is $MDist(o, \bar{o}) = (o - \bar{o})^T S^{-1}(o - \bar{o})$ where S is the covariance matrix协方差矩阵
  - Use the Grubb's test on this measure to detect outliers
- Method 2. Use χ^2 –statistic使用χ^2统计量的多元离群点检测
  - where E_i is the mean of the i-dimension among all objects, and n is the dimensionality维度

$$\chi^2 = \sum_{i=1}^{n} \frac{(o_i - E_i)^2}{E_i}$$

  - If χ^2 –statistic is large, then object o_i is an outlier

Parametric Methods III: Using Mixture of Parametric Distributions使用混合参数分布

- Assuming data generated by a normal distribution could be sometimes overly simplified
- Example (right figure): The objects between the two clusters cannot be captured as outliers since they are close to the estimated mean估计的均值落在这两个簇之间，而不是任何一个簇的内部。这两个簇之间的对象不可能被检测为离群点，因为它们离均值很近



- To overcome this problem, assume the normal data is generated by two normal distributions. For any object o in the data set, the probability that o is generated by the mixture of the two distributions is given by

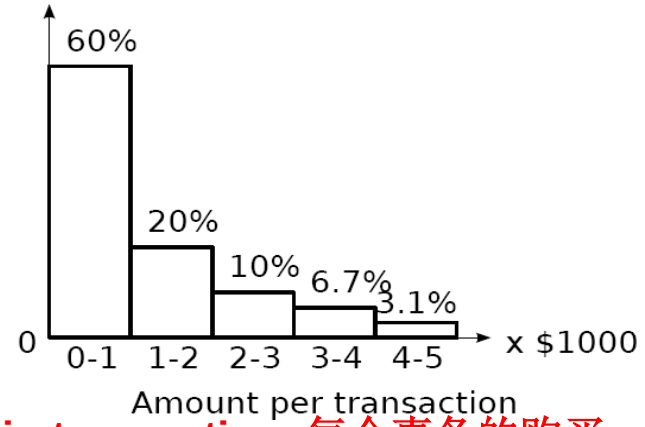
$$Pr(o|\Theta_1, \Theta_2) = f_{\Theta_1}(o) + f_{\Theta_2}(o)$$

where $f_{\theta 1}$ and $f_{\theta 2}$ are the probability density functions of θ_1 and θ_2

- Then use EM algorithm to learn the parameters μ_1, σ_1, μ_2, σ_2 from data
- An object o is an outlier if it does not belong to any cluster

Non-Parametric Methods: Detection Using Histogram
使用直方图检测离群点

- **The model of normal data is learned from the input data without any *a priori* structure.**
- **Often makes fewer assumptions about the data, and thus can be applicable in more scenarios**
- **Outlier detection using histogram:**



  - **Figure shows the histogram of purchase amounts in transactions每个事务的购买量**
  - **A transaction in the amount of $7,500 is an outlier, since only 0.2% transactions have an amount higher than $5,000**
- **Problem: Hard to choose an appropriate bin size合适的箱尺寸 for histogram**
  - **Too small bin size → normal objects in empty/rare bins, false positive将正常点误识别为离群点，导致高假正率和低精度**
  - **Too big bin size → outliers in some frequent bins, false negative将离群点当成正常点，导致高假负率和低召回率**
- **Solution: Adopt kernel density estimation核密度估计 to estimate the probability density distribution of the data. If the estimated density function is high, the object is likely normal. Otherwise, it is likely an outlier.**

离群点检测的统计学方法特性分析

- 离群点检测的统计学方法由数据学习模型，以区别正常的数据对象和离群点。使用统计学方法的一个优点是，离群点检测可以是统计上无可非议的。当然，仅当对数据所做的统计假定满足实际约束时才为真。
- 高维数据的数据分布常常是复杂的，并且很难完全理解。因此，在高维数据上，离群点检测的统计学方法仍然是一个大难题。
- 统计学方法的计算开销依赖于模型。在使用简单的参数模型（如高斯模型〕时，拟合参数通常需要线性时间。当使用更复杂的模型时（如混合模型，那里学习中使用EM 算法）逼近最佳参数值通常需要多次迭代。然而，每次迭代，关于数据集的大小都是线性的。对于核密度估计，模型学习的开销可能高达二次。一旦模型学习成功，每个对象的离群点检测的开销通常都很小。

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Proximity-Based Approaches: Distance-Based vs. Density-Based Outlier Detection基于距离的vs基于密度的离群点检测

- Intuition: Objects that are far away from the others are outliers
- Assumption of proximity-based approach: The proximity of an outlier deviates significantly from that of most of the others in the data set基于邻近性的方法假定：离群点对象与它最近邻的邻近性显著偏离数据集中其他对象与它们近邻之间的邻近性

- Two types of proximity-based outlier detection methods
  - Distance-based outlier detection: An object o is an outlier if its neighborhood does not have enough other points
  - Density-based outlier detection: An object o is an outlier if its density is relatively much lower than that of its neighbors

Distance-Based Outlier Detection

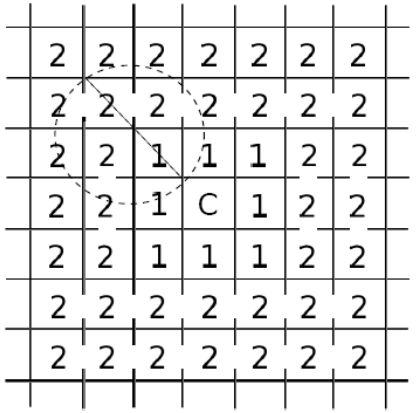
- For each object o, examine the # of other objects in the *r*-neighborhoodof o, where *r* is a user-specified distance threshold （距离阈值r定义的合理邻域）
- An object o is an outlier if most (taking π as a fraction threshold分数阈值，0<π<=1) of the objects in D are far away from o, i.e., not in the r-neighborhood of o
- An object o is a DB(r, π) outlier if $$\frac{\|\{o' | dist(o, o') \leq r\}\|}{\|D\|} \leq \pi$$
- Equivalently, one can check the distance between *o* and its *k*-th nearest neighbor o_k第k个最近邻, where $k = \lceil \pi \|D\| \rceil$
- *o* is an outlier if dist(*o*, o_k) > r，在o的r邻域中，除o之外少于k个对象

- Efficient computation: Nested loop algorithm嵌套循环算法
  - For any object o_i, calculate its distance from other objects, and count the # of other objects in the r-neighborhood.
  - If π·n other objects are within r distance, terminate the inner loop
  - Otherwise, o_i is a DB(r, π) outlier
- Efficiency: Actually CPU time is not O(n^2) but linear to the data set size since for most non-outlier objects, the inner loop terminates early

Distance-Based Outlier Detection: A Grid-Based Method
基于网格的方法

- Why efficiency is still a concern? When the complete set of objects cannot be held into main memory, cost I/O swapping
- The major cost: (1) each object tests against the whole data set, why not only its close neighbor? (2) check objects one by one, why not group by group?
- Grid-based method (CELL): Data space is partitioned into a multi-D grid多维网格. Each cell is a hyper cube超立方体 with diagonal length对角线长度 r/2
- 其中r是一个距离阈值参数，如果有L维，则单元的每个边长为 $\frac{r}{2\sqrt{l}}$

- the level-1 & level 2 cell properties:
  - For any possible point x in cell C and any possible point y such that dist(x,y) ≤ r, y is in a level-1 cell
  - For any possible point x in cell C and any point y such that dist(x,y) ≥ r, y is in a level-2 cell



Distance-Based Outlier Detection: A Grid-Based Method
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- **Pruning剪支** using the level-1 & level 2 cell properties

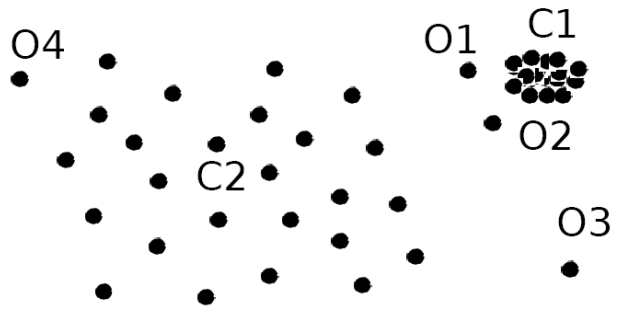
设 a 是单元 C 中的对象数，b_1 是第 1 层单元中的对象总数，b_2 是第 2 层单元中的对象总数。可以使用如下规则：

- 层 −1 单元剪枝规则：根据第 1 层单元的性质，如果 $a + b_1 > \lceil \pi n \rceil$，则 C 中的每个对象 o 都不是 $DB(r, \pi)$ − 离群点，因为 C 和第 1 层单元中的所有对象都在 o 的 r − 邻域中，并且至少有 $\lceil \pi n \rceil$ 个这样的近邻。
- 层 −2 单元剪枝规则：根据第 2 层单元的性质，如果 $a + b_1 + b_2 < \lceil \pi n \rceil + 1$，则 C 中的所有对象都是 $DB(r, \pi)$ − 离群点，因为它们的 r − 邻域中的其他对象都少于 $\lceil \pi n \rceil$ 个。

- Thus we only need to check the objects that cannot be pruned, and even for such an object o, only need to compute the distance between o and the objects in the level-2 cells (since beyond level-2, the distance from o is more than r)

Density-Based Outlier Detection
基于密度的离群点检测

- **Local outliers: Outliers comparing to their local neighborhoods, instead of the global data distribution**
- **In Fig., o_1 and o2 are local outliers to C_1, o_3 is a global outlier, but o_4 is not an outlier. However, proximity-based clustering cannot find o_1 and o_2 are outlier（o1和o2到稠密簇C1的距离小于C2 族中对象到它的最近邻的平均距离，因此，o1和o2都不是基于距离的离群点**



- Intuition (density-based outlier detection): The density around **an outlier** object is **significantly different from** the density around its neighbors
- Method: **Use the relative density of an object against its neighbors as the indicator of the degree of the object being outliers**
- *k-distance* of an object o, $dist_k(o)$: distance between o and its k-th NN
- *k-distance neighborhood* of o, $N_k(o) = \{o' | o' \text{ in } D, dist(o, o') \leq dist_k(o)\}$
  - $N_k(o)$ could be bigger than k since multiple objects may have identical distance to o多个对象到o的距离相等

Local Outlier Factor: LOF

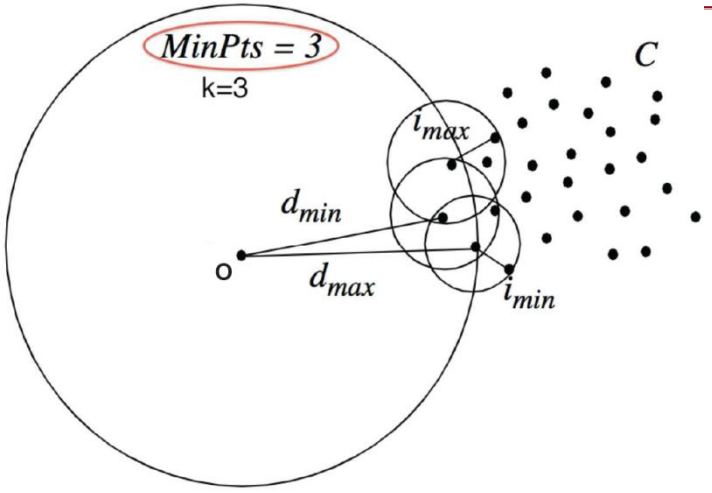
- **Reachability distance可达距离 from *o'* to *o*:**

$$reachdist_k(o \leftarrow o') = \max\{dist_k(o), dist(o, o')\}$$

  - **k is a user-specified parameter控制光滑效果**

- **Local reachability density 局部可达密度of *o*:**

$$lrd_k(o) = \frac{\|N_k(o)\|}{\sum_{o' \in N_k(o)} reachdist_k(o' \leftarrow o)}$$



- **LOF (Local outlier factor局部离群点因子)** of an object o is the average of the ratio of local reachability of *o* and those of *o*'s k-nearest neighbors

$$LOF_k(o) = \frac{\sum_{o' \in N_k(o)} \frac{lrd_k(o')}{lrd_k(o)}}{\|N_k(o)\|} = \sum_{o' \in N_k(o)} lrd_k(o') \cdot \sum_{o' \in N_k(o)} reachdist_k(o' \leftarrow o)$$

- The lower the local reachability density of o, and the higher the local reachability density of the kNN of o, the higher LOF
- This captures a local outlier whose local density is relatively low comparing to the local densities of its kNN

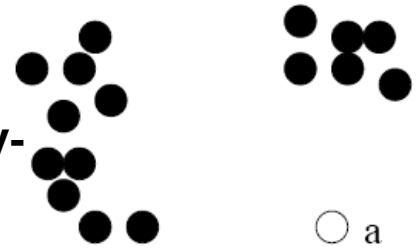
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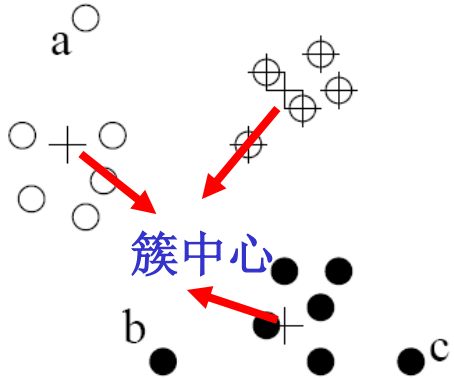
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Clustering-Based Outlier Detection (1 & 2):

Not belong to any cluster, or far from the closest one

- 三种基于聚类的离群点检测An object is an outlier if
  - — it does not belong to any cluster
  - — there is a large distance between the object and its closest cluster
  - — it belongs to a small or sparse cluster
- Case I: Not belong to any cluster
  - Identify animals not part of a flock群: Using a density-based clustering method such as DBSCAN
- Case 2: Far from its closest cluster
  - Using k-means, partition data points of into clusters
  - For each object o, assign an outlier score based on its distance from its closest center
    - avg_dist(co)表示簇中各点到簇中心的平均距离
    - If dist(o, c_o)/avg_dist(c_o) is large, likely an outlier
- Ex. Intrusion detection: Consider the similarity between data points and the clusters in a training data set

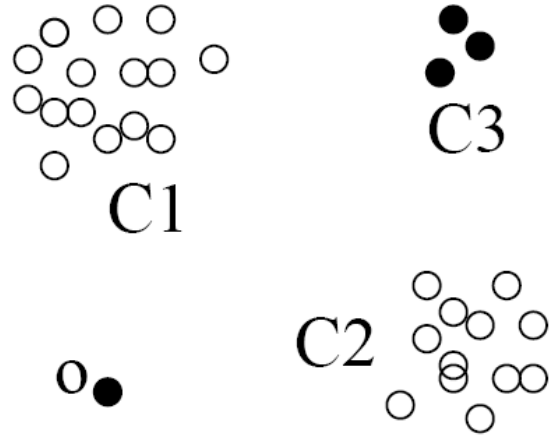




通过考虑训练数据集中的数据点与簇之间的相似性，检测TCP连接数据中的入侵，方法包括三步：（1）使用训练数据集找出正常数据模式，TCP连接数据根据日期分段，在每个段中发现频繁模式。分段中处于多数的频繁模式被视为正常数据模式，并称为“基本连接”。（2）训练数据中包含基本连接的连接被看做无攻击的。这些连接被聚类成簇。（3）把原数据集中数据点与上一步得到的簇进行比较，这些簇的离群点都被看做可能的攻击。

Clustering-Based Outlier Detection (3): Detecting Outliers in Small Clusters

- *FindCBLOF:* Detect outliers in small clusters
  - Find clusters, and sort them in decreasing size降序
  - To each data point, assign a *cluster-based local outlier factor* (CBLOF)基于簇的局部点因子:
  - If obj p belongs to a large cluster, CBLOF = cluster_size X（乘法） similarity between p and cluster
  - If p belongs to a small one, CBLOF = cluster size X similarity betw. p and the closest large cluster



- Ex. In the figure, o is outlier since its closest large cluster is C_1, but the similarity between o and C_1 is small. For any point in C_3, its closest large cluster is C_2 but its similarity from C_2 is low, plus $|C_3| = 3$ is small

Clustering-Based Method: Strength and Weakness

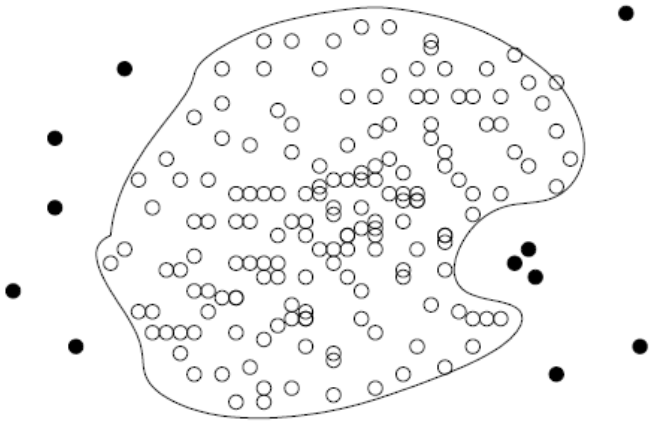
- **Strength**
  - Detect outliers without requiring any labeled data（无监督方式）
  - Work for many types of data
  - Clusters can be regarded as summaries of the data
  - Once the cluster are obtained, need only compare any object against the clusters to determine whether it is an outlier (fast)
- **Weakness**
  - Effectiveness depends highly on the clustering method used—they may not be optimized for outlier detection
  - High computational cost: Need to first find clusters
  - A method to reduce the cost: **Fixed-width clustering**固定宽度聚类
    - A point is assigned to a cluster if the center of the cluster is within a pre-defined distance threshold from the point
    - If a point cannot be assigned to any existing cluster, a new cluster is created and the distance threshold may be learned from the training data under certain conditions

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Classification-Based Method I: One-Class Model

- Idea: Train a classification model that can distinguish “normal” data from outliers
- A brute-force基本的 approach: Consider a training set that contains samples labeled as “normal” and others labeled as “outlier”
  - But, the training set is typically heavily biased高度有偏的: # of “normal” samples likely far exceeds # of outlier samples
  - Cannot detect unseen anomaly([ə nɑməli]异常)



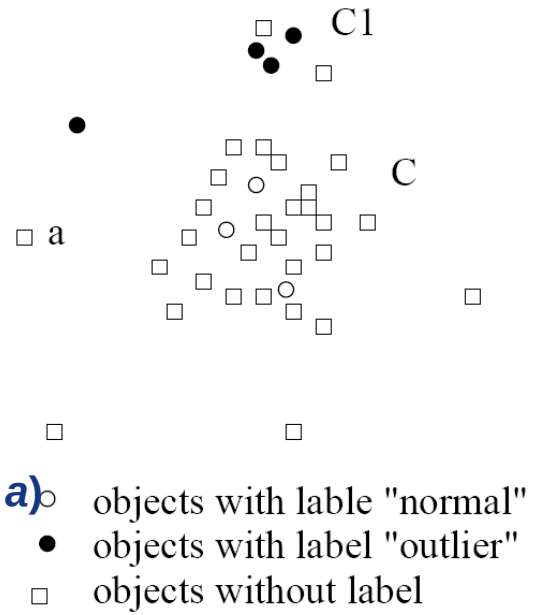
- One-class model一类模型: A classifier is built to describe only the normal class.
  - Learn the decision boundary of the normal class using classification methods such as SVM
  - Any samples that do not belong to the normal class (not within the decision boundary) are declared as outliers
  - Adv: can detect new outliers that may not appear close to any outlier objects in the training set

Extension: 模糊类Normal objects may belong to multiple classes。

为了确定一个实例是否是离群点，可以在该实例上运行每个模型。如果该实例不拟合任何模型，则它被宣布为离群点

Classification-Based Method II: Semi-Supervised Learning

- **Semi-supervised learning: Combining classification-based and clustering-based methods**
- **Method**
  - **Using a clustering-based approach, find a large cluster, C, and a small cluster, C_1**
  - **Since some objects in C carry the label "normal", treat all objects in C as normal**
  - **Use the one-class model of this cluster to identify normal objects in outlier detection**
  - **Since some objects in cluster C_1 carry the label "outlier", declare all objects in C_1 as outliers**
  - **Any object that does not fall into the model for C (such as *a*) is considered an outlier as well**



- Comments on classification-based outlier detection methods
  - Strength: Outlier detection is fast
  - Bottleneck: Quality heavily depends on the availability and quality of the training set, but often difficult to obtain representative and high-quality training data

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Mining Contextual Outliers挖掘情境离群点 I: Transform into Conventional Outlier Detection 转换成传统的离群点检测

- If the contexts can be clearly identified, transform it to conventional outlier detection
  1. Identify the context of the object using the contextual attributes
  2. Calculate the outlier score 离群点得分for the object in the context using a conventional outlier detection method
- Ex. Detect outlier customers in the context of customer groups
  - Contextual attributes: *age group, postal code*
  - Behavioral attributes: *# of trans/yr每年购买次数, annual total trans. Amount 每年购买总量*
- Steps: (1) locate 顾客c's context, (2) compare c with the other customers in the same group, and (3) use a conventional outlier detection method
- If the context contains very few customers, generalize contexts情境一般化 （如：可以假定居住在相同区域、年龄差不多的顾客具有相似的正常行为）
  - Ex. Learn a mixture model U on the contextual attributes, and another mixture model V of the data on the behavior attributes
  - Learn a mapping $p(V_i|U_j)$: the probability that a data object o belonging to cluster U_j on the contextual attributes is generated by cluster Vi on the behavior attributes
  - Outlier score:

$$S(\boldsymbol{o}) = \sum_{U_j} p(\boldsymbol{o} \in U_j) \sum_{V_i} p(\boldsymbol{o} \in V_i) p(V_i|U_j)$$

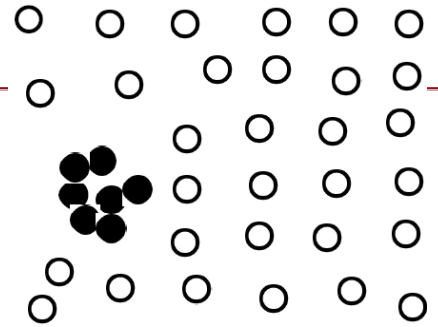
Mining Contextual Outliers挖掘情境离群点 II

Modeling Normal Behavior with Respect to Contexts

关于情境对正常行为建模

- In some applications, one cannot clearly partition the data into contexts
  - Ex. if a customer suddenly purchased a product that is unrelated to those she recently browsed, it is unclear how many products browsed earlier should be considered as the context
- Model the "normal" behavior with respect to contexts
  - Using a training data set, train a model that predicts the expected behavior attribute values with respect to the contextual attribute values
  - An object is a contextual outlier if its behavior attribute values significantly deviate from the values predicted by the model
- Using a prediction model that links the contexts and behavior连接情境和行为的预测模型, these methods avoid the explicit(明确的) identification of specific contexts
- Methods: A number of classification and prediction techniques can be used to build such models, such as regression, Markov Models, and Finite State Automaton有穷状态自动机

Mining Collective Outliers挖掘集体离群点 I
On the Set of "Structured Objects"
图数据上的集体离群点检测



- Collective outlier if objects as a group deviate significantly from the entire data
- Need to examine the *structure* of the data set, i.e, the relationships between multiple data objects

- Each of these structures is inherent to its respective type of data
  - For temporal data (such as time series and sequences), we explore the structures formed by time, which occur in segments of the time series or subsequences
  - For spatial data, explore local areas
  - For graph and network data, we explore subgraphs
- Difference from the contextual outlier detection: the structures are often not explicitly defined, and have to be discovered as part of the outlier detection process.
- Collective outlier detection methods: two categories
  - （1）Reduce the problem to conventional outlier detection
    - Identify *structure units*, treat each structure unit (e.g., subsequence, time series segment, local area, or subgraph) as a data object, and extract features
    - Then outlier detection on the set of "structured objects" constructed as such using the extracted features，集体离群点检测问题就转换成在使用提取的特征构造的“结构化对象”集上的离群点检测

Mining Collective Outliers II: Direct Modeling of the Expected Behavior of Structure Units

- Ex. 1. Detect collective outliers in online social network of customers
  - — Treat each possible subgraph of the network as a structure unit结构单元
  - — Collective outlier: An *outlier subgraph* in the social network一些有悖常理的出现了
    - Small subgraphs that are of very low frequency
    - Large subgraphs that are surprisingly frequent
- （2）Models the expected behavior of structure units directly
- 预先定义结构单元是困难的
- Ex. 2. Detect collective outliers in temporal sequences
  - — Learn a Markov model from the sequences
  - — A subsequence can then be declared as a collective outlier if it significantly deviates from the model

- Collective outlier detection is subtle微妙的 due to the challenge of exploring the structures in data
  - — The exploration typically uses heuristics, and thus may be application dependent
  - — The computational cost is often high due to the sophisticated复杂的 mining process

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Challenges for Outlier Detection in High-Dimensional Data

- 在某些应用中，可能需要检测高维数据中的离群点。维灾难对有效的离群点检测提出了巨大挑战。
  - 随着维度的增加，对象之间的距离可能严重被噪声所左右。也就是说，在高维空间中，两点之间的距离或相似性可能并不反映点之间的实际联系。因此，随着维度的增加，主要使用相似性或密度识别离群点的传统检测方法的效果越来越差。

Challenges for Outlier Detection in High-Dimensional Data

- **Interpretation of outliers离群点的解释**
  - — Detecting outliers without saying why they are outliers is not very useful in high-D due to many features (or dimensions) are involved in a high-dimensional data set
  - — E.g., which subspaces that manifest the outliers or an assessment regarding the "outlier-ness" of the objects （关于对象离群性的评估）
- **Data sparsity数据的稀疏性**
  - — Data in high-D spaces are often sparse
  - — The distance between objects becomes heavily dominated by noise as the dimensionality increases
- **Data subspaces数据子空间**
  - — Adaptive to the subspaces signifying the outliers自适应显示离群点的子空间
  - — Capturing the local behavior of data捕获数据的局部变化
- **Scalable with respect to dimensionality关于维度的可伸缩性**
  - — # of subspaces increases exponentially（指数增加）

Approach I: Extending Conventional Outlier Detection
扩充的传统离群点检测

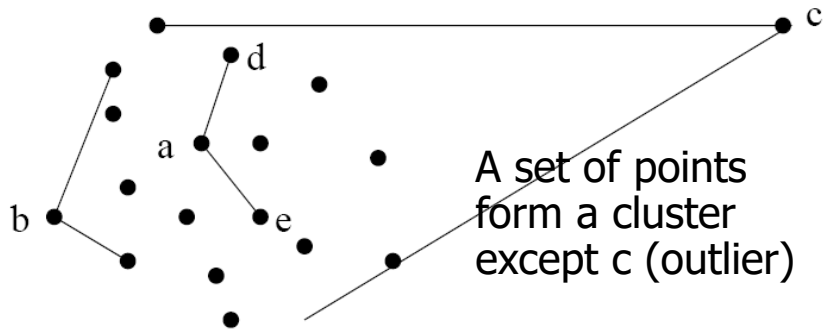
- **Method 1: Detect outliers in the full space, e.g., HilOut Algorithm**
  - Find distance-based outliers, but use the ranks of distance距离秩 instead of the absolute distance in outlier detection
  - For each object *o*, find its k-nearest neighbors: $nn_1(o)$, . . . , $nn_k(o)$
  - The weight of object o: $$w(\boldsymbol{o}) = \sum_{i=1}^{k} dist(\boldsymbol{o}, nn_i(\boldsymbol{o}))$$
  - All objects are ranked in weight-descending order按权重递减序定秩
  - Top-*l* objects in weight are output as outliers (*l*: user-specified parm)
  - Employ space-filling curves空间填充曲线 for approximation: scalable in both time and space w.r.t. data size and dimensionality
- **Method 2: Dimensionality reduction**
  - Works only when in lower-dimensionality, normal instances can still be distinguished from outliers
  - PCA: Heuristically, the principal components with low variance are preferred because, on such dimensions, normal objects are likely close to each other and outliers often deviate from the majority

Approach II: Finding Outliers in Subspaces
发现子空间中的离群点

- Extending conventional outlier detection: Hard for outlier interpretation
- Find outliers in much lower dimensional subspaces: easy to interpret *why* and *to what extent* the object is an outlier
  - E.g., find outlier customers in certain subspace: *average transaction amount* >> *avg.* and *purchase frequency* << avg.
- Ex. A grid-based subspace outlier detection method
  - Project data onto various subspaces to find an area whose density is much lower than average
  - Discretize the data into a grid with φ个 equi-depth等深 (why?) regions
  - Search for regions that are significantly sparse显著稀疏
    - Consider a k-d cube k维立方体: k ranges on k dimensions, with n objects
    - If objects are independently distributed, the expected number of objects falling into a k-dimensional region is $(1/\varphi)^k n = f^k n$, the standard deviation is $\sqrt{f^k(1-f^k)n}$
    - The sparsity coefficient稀疏系数 of cube C: $S(C) = \frac{n(C) - f^k n}{\sqrt{f^k(1-f^k)n}}$
    - If S(C) < 0, C contains less objects than expected
    - The more negative, the sparser C is and the more likely the objects in C are outliers in the subspace

Approach III: Modeling High-Dimensional Outliers
高维离群点建模

- Develop new models for high-dimensional outliers directly
- Avoid proximity measures and adopt new heuristics that **do not deteriorate in high-dimensional data不会在高维数据中退化**



- **Ex. Angle-based outliers基于角的离群点: Kriegel, Schubert, and Zimek [KSZ08]**
- **For each point o, examine the angle ∆xoy for every pair of points x, y.**
  - **Point in the center (e.g., a), the angles formed differ widely**
  - **An outlier (e.g., c), angle variable is substantially smaller**
  - **Use the variance of angles角度的方差 for a point to determine outlier**
- **Combine angles and distance to model outliers**
  - **Use the distance-weighted angle variance as the outlier score**
  - **Angle-based outlier factor (ABOF)基于角度的离群点因子**

$$ABOF(o) = VAR_{x,y \in D, x \neq o, y \neq o} \frac{\langle \overrightarrow{ox}, \overrightarrow{oy} \rangle}{dist(o,x)^2 dist(o,y)^2}$$

  - **Efficient approximation computation method is developed**
  - **It can be generalized to handle arbitrary types of data**

Chapter 9. Outlier Detection

- Outlier and Outlier Analysis 离群点和离群点分析
- Outlier Detection Methods 离群点检测方法
- Statistical Approaches 统计学方法
- Proximity-Base Approaches 基于邻近性的方法
- Clustering-Base Approaches 基于聚类的方法
- Classification Approaches 基于分类的方法
- Mining Contextual and Collective Outliers 挖掘情境离群点和集体利群点
- Outlier Detection in High Dimensional Data 高维数据中的离群点检测
- **Summary** 小结

Summary

- Types of outliers
  - — global, contextual & collective outliers
- Outlier detection
  - — supervised, semi-supervised, or unsupervised
- Statistical (or model-based) approaches
- Proximity-base approaches
- Clustering-base approaches
- Classification approaches
- Mining contextual and collective outliers
- Outlier detection in high dimensional data

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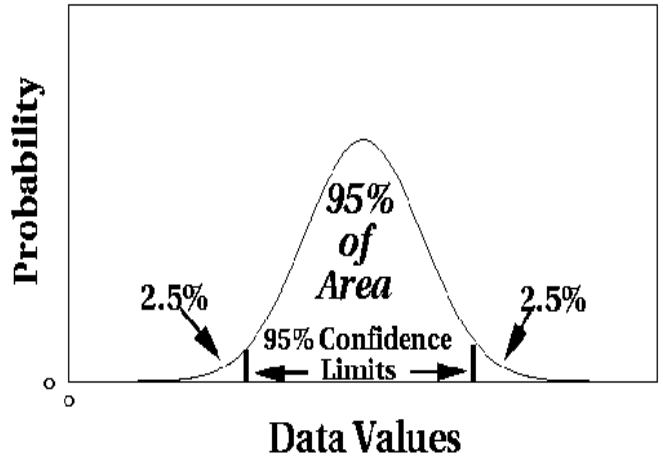
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Outlier Discovery: Statistical Approaches



Assume a model underlying distribution that generates data set (e.g. normal distribution)

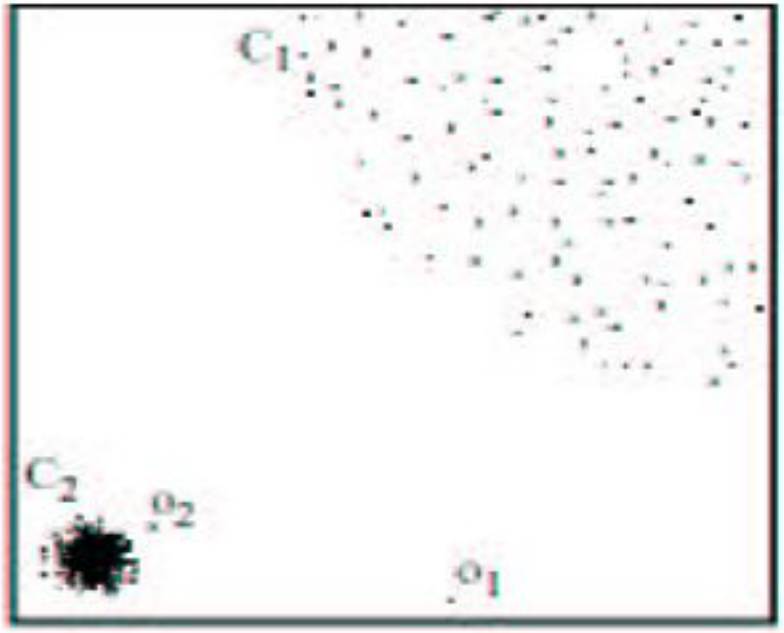
- **Use discordancy tests depending on**
  - **— data distribution**
  - **— distribution parameter (e.g., mean, variance)**
  - **— number of expected outliers**
- **Drawbacks**
  - **— most tests are for single attribute**
  - **— In many cases, data distribution may not be known**

Outlier Discovery: Distance-Based Approach

- Introduced to counter the main limitations imposed by statistical methods
  - We need multi-dimensional analysis without knowing data distribution
- Distance-based outlier: A DB(p, D)-outlier is an object O in a dataset T such that at least a fraction p of the objects in T lies at a distance greater than D from O
- Algorithms for mining distance-based outliers [Knorr & Ng, VLDB'98]
  - Index-based algorithm
  - Nested-loop algorithm
  - Cell-based algorithm

Density-Based Local Outlier Detection

- M. M. Breunig, H.-P. Kriegel, R. Ng, J. Sander. LOF: Identifying Density-Based Local Outliers. SIGMOD 2000.
- Distance-based outlier detection is based on global distance distribution
- It encounters difficulties to identify outliers if data is not uniformly distributed
- Ex. C_1 contains 400 loosely distributed points, C_2 has 100 tightly condensed points, 2 outlier points o_1, o_2
- Distance-based method cannot identify o_2 as an outlier



- Need the concept of local outlier
- Local outlier factor (LOF)
  - Assume outlier is not crisp
  - Each point has a LOF

Outlier Discovery: Deviation-Based Approach

- Identifies outliers by examining the main characteristics of objects in a group
- Objects that "deviate" from this description are considered outliers
- Sequential exception technique
  - simulates the way in which humans can distinguish unusual objects from among a series of supposedly like objects
- OLAP data cube technique
  - uses data cubes to identify regions of anomalies in large multidimensional data

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